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Space and Time in the Brain

> An unmistakable difference exists between spatial and temporal concepts.
>
> —Henrik Lorentz[1]

> [T]ime is an abstraction, at which we arrive by means of changing of things.
>
> —Ernst Mach[2]

> Space and duration are one.
>
> —Edgar Allen Poe[3]

After illustrating and discussing the advantages of the inside-out framework, let's return to the Table of Contents of William James's book (Chapter 1). I hope that I have convinced at least some readers that searching for the brain mechanisms of human-invented terms is not the best way to proceed in neuroscience. But which terms should we abandon or redefine? There are two prominent concepts in James's Table of Contents—space and time—which seem to be non-negotiable, not only in neuroscience but also in everyday life. We can live without sight, sound, smell, or touch, but everything we do seems to occur in space and time. These concepts are built into our language and thinking. We feel ourselves to be distinct from others largely because we cherish our own episodic memories that occurred in a specific place at a specific time. An implicit prediction of this framework is that experiences

1. As cited in Canales (2015).
2. Blackmore (1972).
3. Horgan (2015).

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are broken down into components of what, where, and when, which together can recreate the original episode. Thus, if we want to know ourselves, we must understand not only how the brain stores the "what" but also how the place information and time stamps associated with those experiences are processed, stored, and recalled.

Yet there is something fishy about this postulated triad. In this chapter, I invite you to think about space and time in an open way. You may or may not agree with me, but at least you will experience a different perspective. And if I manage to convince you that time and space are human-invented concepts, perhaps we can begin to think about the rest of the items in James's list similarly: they are constructs of the human brain rather than brain "re-presentations" of real things out there.

Partial list of the Table of Contents of *Principles of Psychology* (William James, 1890)

Chapter XI—Attention
Chapter XV—**Perception of time**
Chapter XVI—Memory
Chapter XVIII—Imagination
Chapter XIX—The perception of "things"
Chapter XX—**The perception of space**
Chapter XXI—The perception of reality

INTERCHANGEABLE SPACE AND TIME IN LANGUAGE

Discussions of space and time often cite the philosopher Immanuel Kant, who argued that these concepts are a priori categories that cannot be studied directly. "Space is nothing but the form of all appearances of outer sense . . . [that] can be given prior to all actual perceptions, and so exist in the mind *a priori*, and . . . can contain, prior to all experience, principles which determine the relations of these objects."[4] Space and time are thus the axioms of the universe, orthogonal to each other and independent from everything else. For episodic memory, such separation also makes a lot of economic sense. If the brain had to store a list of every individual experience of our lifetime (i.e., every combination of *what, where* and *when*), the list would be extraordinarily long. Encoding it would require an extraordinarily large storage capacity, and recalling so many

4. Kant (*Critique of Pure Reason*, p. 71). For Kant, time and space are not concepts but unique preexisting "schemes": only one time and one space in the entire universe. Concepts do not preexist. They are created by some humans and explained to others.

memories would be complicated. Borges's character Funes the Memorius has an impeccable memory and can recall every single moment of his previous day's activity, but it takes him another full day to do that.[5] An alternative solution is to store the what, where, and when components separately and recreate the original episode by re-embedding the what into the where and when. Viewed from this perspective, neuroscience of episodic memory needs to focus on space and time mechanisms in the brain. But do we really know what we are looking for?

The notion of universal space and time in most cultures is used to contrast the vastness and complexities of the universe against the shortness of life. In our everyday conversations, these dimensions are often used interchangeably. In numerous languages, distance and duration are synonymous: "My lab is an hour from home," or "This was a short meeting." The longitudes of the earth are known as *time zones* because high noon occurs at different times in different countries and on different continents.[6] A unit of distance is defined by time: a light-second is the distance traveled by light in 1 s in a vacuum. Today, much of our knowledge of where we are depends on global positioning system (GPS) data, which have no meter metric at all. GPS computes position by determining the time interval that the signal takes to reach the receiver from its various satellites.[7]

Moreover, not all humans share the need for Kant's essential categories. Some cultures have no concept of the passage of time. In fact, almost half of the world's languages do not have grammatical tense (e.g., Mandarin). In all languages examined by linguists, most temporal words have a spatial sense as their primary meaning. The Amondawa, an Amazonian tribe known to the rest of the world only since 1986, lack linguistic structures that relate to time altogether. Not only do they have no word for the abstract concept of time, but they also have no words for the more concrete month or year. The Amondawa do not refer to their ages or celebrate birthdates. Instead, they change their names

5. Borges (1994).

6. Before the cesium atomic clock existed, the official time for both civil and astronomic use was calibrated by the orbital position of the sun and the moon based on a dynamical theory of their motions, called *ephemeris time*.

7. Satellites of the GPS constellation orbit at an altitude of about 20,000 km from the ground with an orbital period of approximately 12 hours. Local time on each satellite is computed by an atomic clock that has 1 ns accuracy. According to general relativity, a clock aboard a satellite should tick faster than a clock on Earth (corresponding to 45 μs advancement per day). If this time advance was not taken into account, the accumulated error would be almost 10 km each day, making the GPS practically useless.

each time they achieve a different status within their community.[8] Similarly, for the Aborigines of inner Australia in the Great Victoria Desert, the concept of Creation exists in the past, present, and future all at once. As they do not believe in tomorrow, they keep no possessions. If food is left from one day to the next, they simply abandon it, even though food is scarce in the desert. Today is today, not a continuation of yesterday. They pass songs or "songlines" from generation to generation to tell of the travels of the creator across the land. The songlines describe the Creator's path, including the location of mountains, waterholes, landmarks, and boundaries, and thus function as a verbal map for navigation.[9] When members of another Australian aboriginal tribe were asked to arrange a set of pictures that illustrated temporal progression (e.g., pictures of a boy, man, old man), they did not arrange the pictures from left to right, as we normally would. Instead, they arranged them from east to west. That is, when they were facing south, the cards were placed from left to right. In contrast, when they faced north, the cards were arranged from right to left. When they faced east, the cards came toward the body and so on.[10] Thus, even without the concept of time, these people can understand ordering, sequences of events, or arrangement, but they do not have a notion of time as independent of all other things or as a context that events occur within.

The indigenous Wintu people, who have lived in the Sacramento Valley, California, do not distinguish personal space from allocentric space. They do not refer to the left or right related to their body scheme but to the river side or mountain side, depending on whether they travel north or south, the two key directions in the valley.[11] In contrast, Finnish people, who live in a flat land with thousands of lakes, have separate words for the cardinal directions (S, E, N, and W = *etelä, itä, pohjoinen*, and *länsi*, respectively) and the intercardinal directions (SE, NE, NW, and SW = *kaakko, koillinen, luode, lounas*, respectively). Far from Finland, members of a small Kuuk Thaayorre aboriginal community on the western edge of Cape York, in northern Australia, also use cardinal directions

8. Of course, this only means that the Amondawa people live in a world of events with sequences, rather than seeing events as being embedded in continuous time (Sinha et al., 2011).

9. Songlines are essentially oral maps of the vast desert, enabling the transmission of navigational skills in cultures that do not have a written language (Wositsky and Harney, 1999).

10. Boroditsky (https://www.edge.org/conversation/lera_boroditsky-how-does-our-language-shape-the-way-we-think).

11. In Wintu culture, the self is continuous with nature; therefore the self never vanishes (Lee, 1950). There are, in fact, mechanisms in the brain that can code for such information. Neurons in the post-subiculum of mice can integrate head direction information with the presence of a wall to the left or right. All we have to decide is what to do with the time that is given us. Thus, the boundary between egocentric and allocentric information is blurred (Peyrache et al., 2017).

even when they talk about personal (egocentric) space: "There is a spider on your east ear." So if you are one of the Kuuk Thaayorre clan members, you need to know which way you are facing to identify the whereabouts of the spider.[12]

These cultural differences suggest that space and time may not be inseparable ingredients of neuronal computation, but instead mental constructions represented only by trained brains. Our concept of memory requires space and time because our thoughts need to be placed into an imagined coordinate system. But these coordinates may exist only in our conceptual world.

If space and time are immaterial, how can they "cause" any change in the brain, which does not have specialized sensors to perceive them? Finding a reasonable answer to such a difficult question first requires a quick tour of space and time as seen by physicists.

SPACE AND TIME IN PHYSICS

Conceptual space and time are dimensionless and unmeasurable so they cannot be studied directly. Modern science transformed these concepts by replacing the abstract philosophical versions with their measurable variants: distance and duration.[13]

Defining Space and Time

The new tools introduced to calculate distance and duration allowed our predecessors to address important questions, such as "Where am I?" [14] and

12. Levinson (2003).

13. Measuring instruments as the yardstick of truth have only been adopted recently. Galileo palpated his pulse and heartbeat to determine a pendulum beat in the tower of Pisa, not the other way around (Canales, 2015)! According to Richard Feynman and colleagues (1963) "What really matters anyway is not how we define time, but how we measure it." However, if we accept that distance and duration can only be interpreted with the help of human-made rods and clocks, we deny that non-human animals can feel and use these dimensions.

14. Mishkin, Ungerleider, and colleagues (1983) distinguished two hierarchically organized and functionally specialized processing pathways: a "ventral stream" from the visual cortex to the inferotemporal cortex for object vision (what) and a "dorsal stream" from the visual cortex to the posterior parietal region for spatial vision (where). Goodale and Milner (1990) pointed out inadequacies in such pathway–function assignment and suggested that instead the ventral stream plays a role in the perceptual identification of objects, while the dorsal stream mediates the required sensorimotor transformations for visually guided actions directed at

"What time is it?" However, the relationship between distance and position, or duration and clock time, is inherently circular. Position is the end point of a displacement, relative to an arbitrarily defined first position. A similar circularity also applies to the instant or "now," which is defined as the end of a duration relative to an arbitrarily defined beginning.[15] Therefore, the question should be rephrased to "What agreed date/time is it?" because time depends on who is measuring it. As I write this paragraph, it is February 17, 2017, by the international calendar, which corresponds to 21 Jumada I 1438 in the Islamic calendar, 22 Shevat 5777 in the Jewish calendar, and 30 Bahman 1395 in the Persian calendar. The calendars show duration from an arbitrary beginning, and thus the definition of "now" depends on the observer's viewpoint.

If space and time require different instruments to measure their quantities, they may have different qualities. In classical physics, the time axis is added to the three dimensions of space. Distance and duration cross each other only at a single point of time and position. This inert and empty theater of space and time allows movement of things within. Special relativity continued in this tradition and always gave the exact location and speed of a particle.[16]

However, even before Newton, physicists recognized that space and time are related to each other through speed, which is the ratio of distance and duration.[17]

such objects. Thus, the function of the dorsal stream is not to extract the *where* but to support object manipulation (*how*).

15. In *Being and Time* (1927/2002), Heidegger wonders "How is this mode of temporalizing of temporality to be interpreted? Is there a way leading from primordial *time* to the meaning of *being*? Does *time* itself reveal itself as the horizon of *being*?" He suggests that the Being (*Dasein* or existence) is not a question about *what*, but about *who*. This existentialist question is reminiscent of the constant inquiry about the observer's role in interpreting the world.

16. The assumption of an exact location of a particle location violates Heisenberg's uncertainty principle. In quantum physics, the instantaneous velocity of a particle is independent of its exact position. Both can never be recorded simultaneously due to the particle-wave equivalence. Any measurement affects either the particle or the wave state. Because waves spread out in space (and time), the phase is meaningless without knowing the frequency. On the other hand, frequency can be determined only over a space (time) span. This uncertainty is also the fundamental source of the debate about the noise versus rhythmic nature of the local field potential or electroencephalogram (EEG) because frequency and time are independent from each other. The presence or absence of an oscillation can be quantitatively determined only over a time span.

17. The "Mertonian mean speed theorem," named after fourteenth-century scholars at Merton College, Oxford, is credited for the modern formula of velocity as the ratio of displacement and duration. However, recently discovered clay tablets of cuneiform writings have indisputably proved that Babylonian astronomers already used geometrical methods as abstract mathematical space and calculated Jupiter's position from the area under a time-velocity graph (Ossendrijver, 2016). To be precise, speed is not distance divided by duration, but by some

Knowledge of any two parameters is sufficient to fully describe the third. As discussed in Chapter 2, when two things are invariably correlated, we often assume that one causes the other or that both have a common cause. Does distance cause duration or vice versa? Or do they have a joint explanation? The independence of space and time in classical physics has been questioned by many. If time is a medium through which things travel, time does not exist without them. To define time, some particles must move across space. Conversely, no event can happen in no time or as Hermann Minkowski, Einstein's teacher, phrased it: "Nobody has ever noticed a place except at a time, or a time except at a place." [18] Time is chronicled by matter.

Time Is an Arrow

In classical thermodynamics, heat flows from hot bodies to cold ones. Every moment is different because collisions of the molecules produce more collisions and therefore friction and heat. The second law of thermodynamics states that the universe moves to increasingly higher disorder, quantified by the term "entropy,"[19] and never backward. According to Arthur Eddington, this one-way flow of entropy is the reason that time moves forward, and it explains the "arrow of time."[20] If time is an arrow, it is a vector, and as such needs to be characterized by two measures: duration and direction. One should notice, though, that increased entropy is only a metaphor and not equal to time or duration. Importantly, using the logic of "entropy explains time," time should move backward in biological systems since in self-organized system entropy decreases.

tangible and measurable change, such as another convenient distance measure or the number of swings of a pendulum. Only when clock time measurements are available does the distance divided by duration formula become valid for speed calculation.

18. As cited in (Barbour, 1999).

19. Entropy—*energy trope* or transformation (*en-trope*)—is the log of the number of quantum states accessible to a system.

20. "The great thing about time is that it goes on" (Eddington, 1928). Eddington is credited for the expression "time arrow." Heraclitus's *panta rhei* refers to a similar idea. However, a recent experiment shows that heat can spontaneously flow from a cold quantum particle to a hotter one under certain conditions (Micadei et al., 2017). For a recent discussion on the direction of time, see Zeh (2002). The directed notion of time makes it a vector, in contrast to the scalar duration. For a discussion and critique on the relationship between entropy and time arrow, see Mackey (1992) and Muller (2016).

The Big Bang theory also supports the time-arrow idea. The Big Bang is often considered as the beginning of everything—the point from which the universe expanded in space and moved forward in time. The theory received a major boost in 2016, when astronomers identified gravitational waves likely originating from a pair of neutron stars or super-massive black holes.[21] However, preexisting matter did not explode within preexisting space. Instead, the Big Bang created everything, including what we call space and time. Thus, there may be no need to consider these terms separately because expansion of space is inherently linked to dilation of time. In contrast, another theory, known as *inflation*, suggests that an exponential process of increases in both space and mass preceded the Big Bang, even if its duration was super short. The Big Bang time may apply only to our part of the universe.[22] Yet another altenative is the universe undergoes an infinite number of Big Bangs with recurring cyclic expansion and contraction of space and time. This view of uncaused eternity is of course less attractive to scholars in search for causal explanations of things with clearcut beginnings.

Spacetime in Physics

In the nineteenth century, the discovery of the constant speed of light and the recognition that light disobeys the principle of causality[23] paved the way to theories of relativity, which rest on an equivalence between distance and duration and the symmetry of time.[24] The classic model was abandoned in favor of a spacetime model of general relativity, where the time axis is a fourth

21. Steinhardt and Turok (2002). A concise account of gravitational field is by Cho (2016).

22. Guth (1997); Tegmark (2014).

23. The experiment by Michelson and Morley (1887), and confirmed by many, refuted "ether" as a medium through which light and electromagnetic fields travel. They showed that the speed of light (~300,000 km/s in a vacuum) is the same whether it is measured in the direction of the Earth's motion or at a right angle to it. Light thus has special qualities; its speed is not accelerated or decelerated by other things moving.

24. Hermann Minkowski formulated relativity theory, in which space and time were coordinates in an abstract four-dimensional "spacetime"; the time coordinate was imaginary. In Minkowski's spacetime, relative times and spaces appear as projections relative to an observer. "Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality" (Minkowski, 1909). Prior to Minkowski, spacetime unity was already postulated in a literary form by H. G. Wells in his *The Time Machine* novel: "There are really four dimensions, three of which we call the three planes of Space, and a fourth, Time."

dimension. Einstein postulated that the time of an event affects its location, and vice versa,[25] so that distance and duration cannot be measured independently from each other.

All major theories of twentieth-century physics have been based on the spacetime continuum model. There is no change in spacetime; it does not have a past, present, or future. Spacetime of the universe is like a movie that contains the entire story. But we are looking at the universe through a narrow slit, giving the illusion that what we are watching through the slit is the now. What we recall from memory is the past, and what we attempt to deduce from the past is the future (see Figure 2.1). Thus, from the human observer's point of view, the world is in perpetual motion, while from the view of a very distant observer, nothing changes. Conversely, the Earth is not moving from my current perspective, but it does when I view it from a spaceship. In general relativity, time travel is symmetric. Einstein famously declared that "the separation between the past, present, and future, holds nothing more than a persistent, stubborn illusion, however strong it may be."[26]

Time is reversible: it can be compressed or expanded. It can be perfectly described by clocks without a conscious human brain to sustain it.[27] There is a catch here, though. Clocks neither make time nor represent it. Time does not mean anything to a clock. Ticking has no abstract meaning without an observer.

If time always moves in one direction, but a particle can return to its original place, displacement has no temporal correspondence. That is, no event can happen in zero time. A photon can never rest because at rest it would have zero energy, and so it would not exist. In relativity theory, past and future are completely symmetric, and thus time is scalar. The concept of "now" is irrelevant. In fact, the theory would still hold if the planets revolved in the opposite direction or if the universe went backward. Many interesting things can happen when time moves backward. A positron becomes an electron moving backward in time, as postulated by Richard Feynman, and its positive charge is therefore only an illusion due to negative time.[28]

25. Einstein (1989, 1997). [NeuroNote: Einstein's son Eduard was schizophrenic.] Many excellent books are available about the relationship between space and time and the implications of the spacetime concept (Canales, 2015; Muller, 2016; Rovelli, 2016; Weatherall, 2016; Carroll, 2000).

26. While Einstein put this statement in writing, it was not in a scientific publication but in a personal letter sent to Michele Besso's sister after Besso's death. In the theory of relativity, light is the grounding element of truth, to which everything relates secondarily.

27. I assume that Einstein made this statement out of frustration when he realized that his audience did not comprehend his new spacetime formulation. In the everyday, practical world, clock time tells you what you need for your regular business.

28. In general relativity theory not only the flow of time is an illusion, but change itself as well. Several excellent books discuss these interesting concepts, including the idea that the concept

Is Time Redundant?

While the theory of relativity and quantum mechanics are both internally consistent and beautiful, they contradict each other.[29] The heart of the dispute is whether the world is continuous or consists of discrete quanta. Several attempts have been made to combine these vast fields of physics, most recently in the theory of loop quantum gravity.[30] Here the grains of space make up discontinuous space. This resonates well with the comment of French philosopher Henri Bergson, "When the mathematician calculates the future state of a system at the end of time *t*, there is nothing to prevent him from supposing that the universe vanishes from this moment till that, and suddenly reappears."[31] For Bergson, time is not something out there, separate from those who perceive it. However, the grains of space are not simply quanta, and they are not in space, as the quanta themselves make up space by their sheer interactions. The theory that describes the interaction of space and matter does not contain time, even though everything is in continual motion.

The apparent contradiction between physics and the feeling of the flow of time has upset many people, including the German philosopher Martin Heidegger, who concluded that physics cannot reach the most fundamental aspects of reality. Equations are simply symbols, but reality may be different. "Once time has been defined as clock time, then there is no hope of ever arriving at its original meaning again."[32] But the problem is perhaps different from what Heidegger

of time is not needed to describe the physics of the world (Toulmin and Goodfield (1982); Barbour, 1999; Carroll, 2000; Greene, 2011; Tegmark, 2014; Rovelli, 2016). In contrast, Lee Smolin (2013) forcefully argues that the most essential feature of the universe is time, and physics took the wrong path when it fused it with space. Smolin believes that the laws of physics constantly evolve in time. Gravity may act differently at different points in time, so the past, present, and future can be still distinguished. We may still be surprised about what happens in the next moment.

29. To know that duration is *time*, intervals need to be calibrated against something else. To calibrate the duration between the swings of a pendulum, another instrument is needed (e.g., a faster pendulum or clock with ticks), and so on. Even the fastest atomic clocks still measure time by quanta. According to quantum theory, spacetime is discrete. With clocks, we do not measure time per se but instead count quanta and multiply them by duration. But duration is measured by counting faster quanta, and so it goes *ad infinitum*.

30. Smolin (2013); Rovelli (2016).

31. Bergson (1922/1999); Canales (2015).

32. Heidegger (1927/2002). [NeuroNote: Heidegger had a major "nervous breakdown" after World War II, possibly due to his guilt for aligning with the Nazi regime combined with his existential uncertainty.]

thought, and physics alone may not be able to provide answers to the space and time issue. Instead, it is up to neuroscience to address them.

SPACE AND TIME IN THE BRAIN: REPRESENTATION OR CONSTRUCTION?

So far, these discussions within physics and between physics and philosophy have had little impact on neuroscience. Instead, Newtonian space and time are present in every nook and cranny of neuroscience. [33] Research on space and time in the brain has generated two independent fields with separate literatures while largely ignoring the possibility of space-time equivalence. Neuroscience researchers continue to perform experiments in the framework of classical physics using practical rod and clock measurements. There is no problem with such an approach, as long as we are aware that our measurements are based on human-designed instruments. But it is entirely different to assume that correlations with such metrics can illuminate the concepts of space and time.

The Grounding Problem of Space and Time

An alternative to taking physics seriously is to question whether the laws of physics apply to "psychological" time. After all, the math that describes those laws is also a product of human thought.[34] Math is an axiomatic system, always starting with some assumptions to build internally coherent and beautiful theories. Perhaps physics is just a type of science that has laws that are different from biology. After all, increasing entropy is the rule of the inorganic world, whereas decreasing local entropy is the norm in biology. Prediction or even determinism rules the nonliving world, whereas prediction becomes an increasingly more complex problem in biology.

33. [NeuroNote: Isaac Newton had what today we would call bipolar personality with grandiose delusions at the end of his life. He thought he was appointed by God to describe truth to the world.

34. This argument can be fueled further by Kurt Gödel's theorem, which traumatized mathematicians and scientists alike: all mathematical theories are incomplete or, in its milder form, any theorem or set of definitions is incomplete. Mathematical truth cannot be challenged through the use of logic. Gödel's argument can be also extended to neuroscience. No branch of science is superior to another. [NeuroNote: Gödel suffered from bipolar disorder, as did the physicist Ludwig Boltzmann, the mathematician Georg Cantor, and many other truly great minds of science.]

A related argument against "physicalism," one often used by philosophers, is that the world does not come with built-in units of distance and duration. Human-invented instruments do not create space or time, nor do they measure them. A rod is just a rod. A clock just ticks, and its ticking moments must be compared to some other change, such as the distance traveled by light in a given time unit. This comparison or calibration needs be made by a third party—a human. Because humans define the units of rods and clocks, this process inevitably defines our notion of space and time.

The problem of the human observer and the need for grounding are analogous to the problems in neuroscience that we discussed in Chapters 1 and 3.[35]

There appears to be a vast gap between the classical or macroscopic world that we live in and the microscopic world ruled by relativity and quantum physics. We can ignore this gap, or we can try to examine the sources of the apparent contractions and the boundaries between the macroscopic and microscopic.[36]

Whatever path we take, we need to face at least two issues. First, we must recognize the difference between the spatial and temporal properties of the *process of representing* something and the *representations* of space and time. Even if neuronal activity is reliably correlated with the spatiotemporal sequence of events (measured against a rod or timer), such correlations do not mean that neuronal activity computes distance or duration per se.[37] In other words, we must not conflate the description of events with their subjective interpretation (Chapter 2). Second, we need to clarify whether the *time* we talk about in neuroscience experiments corresponds to the time of the universe (physics), the "lived time" of philosophy or "practical time" as measured by the clock. The same scrutiny should apply to space as well. The essence of the problem is that the human brain interprets both the meaning of its responses, as measured against instruments, and the meaning of the units measured by instruments without independent grounding. Measuring relative space and time via rods and clocks is essential for relating and contrasting experiments across laboratories and has yielded enormous progress in neuroscience. Yet it does not get to the heart of the problem: What do space and time mean for the brain?

35. Before precise instruments to measure time were available, people thought that perception and memory were instantaneous. Indeed, without a comparison to something else, no counterargument could have been made because it feels like no delays occur.

36. The bestseller by Rosenblum and Kuttner (2008) discusses many aspects of quantum physics and its implications for a variety of complex issues, including human consciousness. Unfortunately, nearly everything I have read on the complex issues of physics eventually looks for solutions in God, consciousness, and the like, rather than examining the sources of inconsistencies between internally consistent theories.

37. Dennett and Kinsbourne (1992).

SPACE IN THE WORLD VERSUS SPACE IN THE BRAIN

The brain mechanisms for measuring space and time are inevitably inferential because, while we can sense sight, sound, odor, touch, or our movements through specialized receptors, we have no sensors for space and time. In other words, with the outside-in approach, we cannot track down its existence. Even so, neuroscientists initially defined space from an egocentric perspective that reflects how our sensors feel the world out there. This approach led to the view that many different realms of space exist depending on the perceptual system in use. When the body and head are fixed, but the eyes can move, the space is the visual field itself, called *visuo-ocular motor space*. When head movement is allowed, we encounter the *visuo-cephalo motor space*. Locomotion leads to *visuo-locomotor space*. Even visual-ocular motor space can be divided into *foveal* and *peripheral fields*. Similar divisions were applied to auditory, olfactory, and somatosensory space. An object in the hand sits in *palm space*. A candy in your mouth is in *oral space*. The things within range of an arm movement correspond to *reach space. Instrumental space* can be defined as things we perceive through instruments. When riding in a car, the bump we feel when driving through a pothole is referenced to outside the car. We can expand our visual space by orders of magnitude by looking through a telescope.

Although the multitude of space terms in these early experiments may appear bizarre, pioneers of egocentric space exploration provided experimental support for such ideas and found brain-damaged patients with selective deficits in several of these categories. When children were asked to identify mirror-image objects by using both hands without vision, performance was worse when their arms and wrists were immobilized so that only the fingers could move, compared to another condition when the arms and wrists were free to move but the fingers could not perform multidigital touch. The interpretation is that identifying asymmetry or symmetry is easier when the perception can be grounded to another axis; in this example, the hand position in reaching space relative to the postural body space. Similarly, congenitally blind children learning bimanual Braille reading are more confused by patterns that are mirror images of the other Braille patterns than are sighted children of the same age. Possibly, visual information aids the placement of palpated objects in both tactile space and body-centered space.[38]

38. From numerous experiments, Jacques Paillard concluded that mental "maps" of the external world are constructed by the body's own movements (Paillard, 1991). The experiments using blind children were performed by Martinez (1971). This framework was perhaps driven by Nicolai Bernstein's model suggesting that movement is not driven by neuronal representation of individual of joints and muscles but instead guided by space representation (Bernstein, 1947/1967).

Patients with posterior parietal cortex damage illuminate the difference between object and space perception. These patients have difficulty in following a moving object, pointing or reaching to a visible target, learning routes, and recognizing spatial relationships, but they have relatively few deficits in navigation. The personal space effects are more dramatic when the damage affects the right hemisphere than the left hemisphere. These individuals neglect their contralateral body space, shaving only one side of their face or finishing food only from one side of the plate. If asked to copy a picture, such as a clock, they often draw only one half. When a well-studied patient in Milan, Italy, was asked to imagine himself facing the Piazza Del Duomo and describe the scene, he correctly identified buildings on his right but could not recollect things on the left. When he was asked to imagine standing at the opposite end of the Piazza, he listed the buildings and structures on the other, previously neglected, side, which was now to his right.[39] Such "hemi-neglect" patients can perceive and recall objects per se but are unable to access them from imagery or describe them in proper geometrical relationship. An image on the retina is not sufficient to perceive space. The brain must also know where the eyes and head are pointing.[40]

Physiological experiments in monkeys corroborate these observations in humans. The posterior parietal cortex and its partners form a body-centered coordinate system that encodes spatial relationships between objects and the body.[41] Population firing patterns of neurons in the parietal area combine inputs from the environment and various parts of the body. However, this area does not contain a topographic map of either the body or the environment.

39. Holmes (1918) described deficits in spatial orientation in soldiers after World War I with penetrating missile wounds in the parietal lobe. The paper by Bisiach and Luzzatti (1978) was the first to describe that patients with right posterior parietal damage not only ignore things to their left in their vision but also in their imagery. See also van den Bos and Jeannerod (2002).

40. Organizing hand trajectories and other multijoint movements is a super-complex task, yet we do it without effort. *Trajectory* refers to the configuration of hand in space from its initial to final position. Giving distance and duration commands to multiple muscles on the fly would require extraordinary computations. Instead, it seems that a higher order plan of geometric simulation guides movement. Signatures written on a very small piece of paper or on a large board by the same individual are scaled versions of each other. This fact illustrates that trajectory planning and execution are size invariant, which would be difficult to achieve with distance–duration transformations only and without a normalization process (Flash and Hogan, 1985).

41. Numerous psychophysical observations and clinical data with parietal cortex–damaged hemineglect patients had been difficult to reconcile under the assumption that a fixed coordinate frame is used for object localization. However, when positions of objects are represented in multiple reference frames simultaneously in a model of parietal neurons, the inconsistencies diminish (Pouget and Sejnowski, 1997a, 1997b).

Instead, the main job of this network is to integrate such information from the many inputs it receives, several of which do have topographic maps.[42] Such brain areas include the premotor cortex and the thalamic nucleus putamen, where neurons represent visual space near the face in "head-centered" coordinates and the space near the arm in "arm-centered" coordinates.

The parietal cortex also has prominent connections with the dorsolateral prefrontal cortex. The so-called memory fields in the prefrontal region are topographically organized and respond when the eyes move to a remembered position in two-dimensional space. This idea is beautifully illustrated by the induction of a "memory hole" in a circumscribed area of the remembered visual field by a small cortical lesion. Objects presented in the visual field without parietal-prefrontal cortical representation are seen, but they cannot guide the eye to focus on them. Overall, these and many other experiments[43] show how sensory inputs are converted not just into two-dimensional scenes and isolated objects, but also into relationships between the objects and the body. Such mappings can be acquired only via active movements to ground the relationships.[44]

A general principle that emerges from physiological experiments is that the brain does not encode any general and uniform space somewhere within its networks. Instead of a single master coordinate system, most behaviors require multiple coordinate systems and spatial structures represented relationally in different brain regions in varying formats.[45] The next issue to address is how

42. Multimodal signals are combined in the lateral intraparietal cortex (LIP) and area 7a of the posterior parietal cortex. The integration of visual, somatosensory, auditory, and vestibular signals can represent the locations of stimuli with respect to the observer and within the environment. The dorsal medial superior temporal area (MSTd) combines visual motion signals with eye movement and vestibular signals. This integration is believed to be critical for specifying the path on which the observer is moving (Andersen, 1997). Many LIP neurons fire in anticipation of eye movements and are part of the corollary loop of movement planning (Chapter 3).

43. These short paragraphs did not do justice to this very large area of active research. For reviews, see Goldman-Rakic et al. (1990) and Gross and Graziano (1995).

44. Transformation may be symmetric when properties remain unchanged, such mirror symmetry, rotational symmetry (rotating the coordinated), and translational symmetry (shifting the coordinates), or asymmetric ("symmetry breaking") when the transformation produces new qualities; for example, the transformation between sensory cues and an abstract feature, such as a name associated with a face. The "Jennifer Aniston" or "grandmother cells" in the hippocampus (Quian Quiroga et al., 2005) have nothing to do with the intrinsic properties of those neurons (they are just regular pyramidal cells), but their property arises from their complex relationship to other neurons. For a general model of transformations, see Ballard (2015). Properties thus reflect mainly relationships.

45. Graziano et al. (1994).

eye- and body-centered (egocentric) spatial representation is transformed into a world-centered (allocentric) coordinate system.

Navigation in Space and the Hippocampal System

Another major target of the posterior parietal region is the parahippocampal cortex, which projects to the hippocampus by way of the entorhinal cortex. The computations of hippocampal circuits primarily relate to spatial navigation. Hippocampal and entorhinal principal neurons have place fields and grid fields, which explicitly define the *x-y* coordinates of a two-dimensional environment to generate a map (see Chapters 5 and 7). The positions defined by these cells collectively provide an allocentric, nonvectorial representation of space. This spatial map theory, proposed by John O'Keefe and Lynn Nadel, was inspired by Kantian philosophy.[46] In this model, space is not an independent entity out there; the brain constructs it from parts and relationships. How does the brain do this, given that there are no dedicated sensors for space per se? Distance and duration are not derived from first principles. Instead vision, hearing, olfaction, and proprioception are used to deduce the location of and distances to objects. As the animal moves through the environment, the map is formed by multiple mechanisms, including counting the number of steps, self-motion–dependent optic and somatosensory signals, and vestibular acceleration signals.[47]

46. Kant (1871). In *The Hippocampus as a Cognitive Map*, O'Keefe and Nadel (1978) discuss Kant's view of space and how Kant's view relates to allocentric definition of space (pp. 23–24).

47. Various combinations of activity in the otolith organs and semicircular canals uniquely encode every translation and rotation of the body (Angelaki and Cullen, 2008). This information is complemented by signals from the retina. A subset of ganglion cells, the direction-sensitive ganglion cells, is specialized for detecting image motion, called *optic flow*. These neurons mainly encode two motion axes of high behavioral preference: the body axis and the gravitational axis. Translation movement ("direction of heading") induces optic flow that diverges from a point in space around the animal and follows lines of longitude in global visual space. Rotary optic flow follows lines of latitude, circulating around a point in visual space. Subtypes of direction-sensitive ganglion cells align their preferences with one of four cardinal optic flow fields and encode the gravitational and body axes, which are decoded by the brain into translational or rotational components. Binocular ensembles form a panoramic view, and their combinatorial activity signals when the animal rises, falls, advances, or retreats (Sabbah et al., 2017). This system is hard-wired and works hand in hand with the vestibular system. Motion-induced somatosensory stimulation, such as whisker activation by walls, also produces valuable signals, called *haptic flow*. The vestibular signals compensate for eye and head movements to stabilize the retinal image. Discrepancy between visual and vestibular signals induces motion sickness in a car, airplane, train, roller coaster, or even a virtual reality simulation.

TIME IN THE WORLD AND IN THE BRAIN

Our travels and memory recall feel as though they involve time.[48] Is time really critical for brain computations? Many experiments have explored the brain mechanisms that may support timing. We often argue that timing is critical to survival in the real world. A well-timed extension of the paw may mean lunch for a predator, and a quick jump at the right moment may save the life of its prey.[49] Estimation of elapsed time (duration) is essential for anticipating events, expecting reward, preparing decisions, planning actions, and structuring working memory. Time researchers often make a distinction between perceptual-motor timing at the subsecond scale and cognitively mediated interval detection at the suprasecond scale. They associate these scales with mechanisms supported by the cerebellum for short durations and networks in the basal ganglia and in the prefrontal, motor, and parietal cortical regions for longer durations.

Three procedures are used to investigate the passage of time: estimating an event duration, producing intervals, and reproducing intervals, as in a syncopation task. In animal experiments, the experimenter programs the apparatus so that reward is available at, say, 10-s intervals. The participant then reports the learned rule by strategically concentrating responses near the end of the target interval. Many animals are good at this game, giving the impression that their nervous systems have specialized time-keeping mechanisms. After sufficient training, the mean response intervals will correspond to the target intervals. The dispersion of the responses around the mean is proportional to the criterion duration in all species investigated: short interval targets evoke small errors, whereas longer target intervals evoke proportionally larger errors. The

48. I expect that the immediate reaction of the reader is to say that "integration occurs over time." Yet, no matter how natural this link feels, integration is a sequential operation and is just the same without time. Mathematical integration does not need a temporal factor. Physical instantiation of integration needs succession but no time. Integration can occur quickly or slowly, but this speed parameter is not time per se either and can be substituted by the engineering term "gain" or rate of change that can accelerate or decelerate sequences (Chapter 11). Several chapters of Hoerl and McCormack (2001) contain interesting discussions about the relationship between time and memory.

49. Gallistel and Gibbon (2000) suggest that computation of time by some hypothetical brain circuit represents an elementary aspect of cognition. Alternatively, brain rhythms can serve the same purpose, temporally coordinating neuronal events across structures, without dedicating timekeeping mechanisms (Buzsáki, 2006). However, all neurons that take part in brain oscillations compute other functions as well. Here it is important to reiterate the broad definition of computation, which is calculating and describing the relations between algorithmic steps.

lawful relationship between the mean and the distribution of error responses, quantified by standard deviation, is termed the *scalar property* of interval timing mechanisms. What brain mechanisms are in charge to calculate intervals?

Timekeepers in the Brain?

With analogy to ticking clocks and hourglass sand timers, two different brain mechanisms have been assumed to track time: *neuronal clocks* and *ramping timekeepers*. A candidate mechanism for neural clocks is brain rhythms, which can produce relatively discrete "ticks" with a frequency span over several orders of magnitude (Figure 10.1). An example ramp mechanism is the accumulation or integration of spikes of many individual neurons. Integrating spikes over time is analogous to sand flowing through the hourglass. The integrator can be reset at any point (i.e., at preselected threshold), a command is forwarded to a response system, and the integration can start all over again. Thus, while the

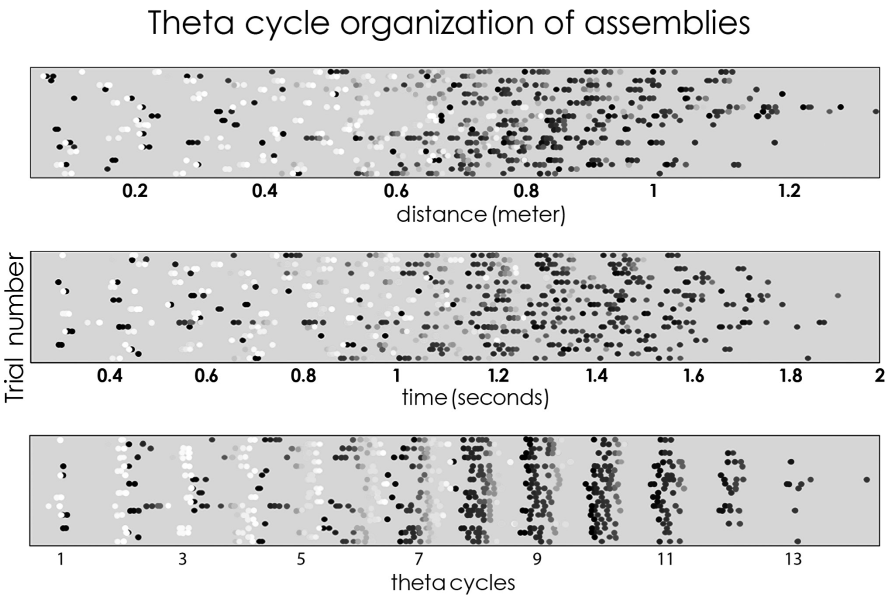

Figure 10.1. Clock time versus rate-of-change in the brain. Place cell assemblies in the hippocampus are organized by theta oscillations ("brain time"). Each raw of dots is a trial of the spiking activity of ten place cells (indicated by different shades of gray). *Top panel*: Trials are shown against distance of the maze. *Middle panel*: Trials are shown against elapsed time from start. *Bottom panel*: Trials are shown as phase-locked activity of neurons in successive theta cycles.
Courtesy of Eva Pastalkova and Carina Curto.

integrator model can have many real intervals as measured by the accumulated sand, the clock has discrete ticks and only virtual intervals between the ticks. With some variation, all timekeepers follow these basic ideas. Duration estimation based on counting spikes deteriorates over time so that the estimated error at any time point is proportional to the elapsed duration since the beginning of the integration (Chapter 12).[50]

Where are the timers in the brain? This question has been asked repeatedly, and two solutions have been offered. An early idea was a central clock that distributes ticks to various brain regions.[51] However, no such brain command system has been found for intervals of milliseconds to minutes.[52] A more contemporary idea is that time is created in each brain system[53] according to local needs. Thus, time is everywhere but calibrated locally and coordinated across networks when needed. As discussed in Chapter 7, any network that can support the self-sustained, sequential activation of neuronal assemblies can potentially track the passage of time. Yet, the local timer model has its own problems. Physiological, pharmacological, and lesion experiments inevitably face the

50. I am aware that these short sentences reflect only a telegraphic summary of the large literature on the psychometric and neuroscience of timing behavior. Luckily, several excellent review papers are available for the interested reader: for example, Church (1984), Michon (1985); Gibbon et al. (1997); Buonomano and Karmarkar (2002); Mauk and Buonomano (2004); Ivry and Spencer (2004); Nobre and O'Reilly (2004); Buhusi and Meck (2005); Staddon (2005); Radua et al. (2014); Mita et al. (2009); Shankar and Howard (2012); Howard et al. (2014); Buonomano (2017).

51. Church (1984). A recent update to these internal clock ideas is that the striatum provides the beats and distributes them to the rest of the brain as needed (Matell and Meck, 2004). However, it is unlikely that the brain has a universal master clock. As in physics, time is a ubiquitous, localized phenomenon for each structure and function in the brain, referenced to some other measure (such as clock time). Time is a relation of events.

52. Neuronal mechanisms that track time at minutes-to-hours scales are scarce. It is believed that such long timing relies on external mechanisms, such as hunger, sleepiness, light, and temperature.

53. In the human EEG literature, the best-known correlate of such time accumulation mechanisms is the *contingent negative variation* (CNV; Walter et al., 1964) and its *Bereitschaftspotential* kin (Kornhuber and Deecke, 1965) that builds up with maximal amplitude above the supplementary motor area seconds before a voluntary action. Because these electrical patterns begin to emerge before the subject is aware of moving, they triggered an intense discussion about free will (see, e.g., Libet, 1985). Error correction in the hippocampus is achieved by the distance–time compression mechanisms of theta oscillations. In each theta cycle, segments of navigation distance and segments of memory and planning are replayed repeatedly and in an overlapping manner. Approximately seven items are repeatedly replayed in multiple theta cycles, and each subsequent cycle gains and loses only one new and one old item, respectively (Lisman and Idiart, 1995; Dragoi and Buzsáki, 2006; Diba and Buzsáki, 2008). This is an effective error-correcting safety mechanism.

problem of dissociating the temporal sequences of representations from the representation of time or duration per se. If behavioral sequences and time are so intricately intertwined that one can perfectly predict the other, why do behavioral sequences need a time label? Let us examine how behaviors are modified under external clock control.

Behavioral Mediation of Time

In interval-timing experiments, animals are always doing something. They figure out ways to pass time by generating stereotypic behaviors. Rodents superstitiously circle around, scratch the wall or the food dispenser, rear repeatedly, or otherwise occupy themselves to avoid a premature press of the lever, especially if that action comes with a penalty. If animals are given the opportunity to run in a wheel, dig in wood shavings, climb ladders, or crawl through tunnels, their timing performance drastically improves compared to an environment where there is not much to do.[54] Humans are not very different. We scratch our heads, bite our nails, or sip coffee while waiting for an idea to pop up in our brains. When in line for a table in a restaurant, we engage in conversations with strangers, check our i-Phones, or impatiently bug the host. Just watch teenagers rhythmically move their legs in the doctor's office while waiting for their appointments. Timekeeping involves doing.

"Time Cells" in Navigation Memory Systems

Without a ruler, distance can be calculated by a time-generating mechanism combined with instantaneous travel velocity, as in the GPS system. All ingredients needed to calculate a route, including speed, duration, and direction, are present in the hippocampal-entorhinal system. Many neurons are modulated by velocity in these structures, where directional input is provided by the head direction system (Chapter 5).

Elapsed duration can be tracked by the accumulation of neuronal spikes in a cell assembly sequence relative to a time-measuring instrument. For example, the passage of time is faithfully recorded by internally evolving neuronal trajectories that maintain information about past memories and planned goals without spatial displacement of the animal (Figure 7.6). The duration estimation error from such cell assembly sequences does not increase proportionally

54. Staddon and Simmelhag (1971).

with elapsed time, but stays relatively asymptotic even after delays of tens of seconds.[55] Based on activity from a hundred neurons in the hippocampus, the accumulated error after 15 s can be less than 20%. This timekeeper may not sound very impressive, at least compared to clocks that have been perfected by humans for centuries, but by increasing the number of recorded neurons, the estimation error can be substantially reduced. Moreover, such precision is good enough for the brain because errors of this size are well-tolerated when computation spreads over tens of seconds.

Why would timekeeping be an important function of the hippocampal system? The typical answer is that the hippocampus and its partners are in charge of our episodic memories, which by definition must be placed into space and time. The postulated need for temporal embedding of episodic memories and the ability of cell assemblies to track time effectively prompted Howard Eichenbaum at Boston University to dub hippocampal and entorhinal neurons "time cells."[56] His group designed a series of experiments to dissociate duration coding from distance coding. They trained rats to run on a treadmill for a target duration or a target distance in a hippocampus-dependent memory task. Most hippocampal and entorhinal neurons responded to both time and distance; that is, they fired reliably and repeatedly on subsequent trials at the same distance or at the same time from the beginning of the run. A minority of neurons was relatively selective for time spent on the treadmill, while the activity of an equally small fraction was better correlated with distance. These results led to the hypothesis that the neurons whose activity correlates with duration may be the key missing ingredient of episodic memory.[57] Investigating the relationship between coding for distance and coding for duration is important because such experiments may allow neuroscientists to confront the concepts of space and time in the context of neuronal mechanisms. However, the more we scrutinize time and its alleged brain mechanisms, the more puzzles arise.

It is hard to justify the hypothesis that there are neuronal networks whose sole function is to clock time. Even when overt motor correlates can be factored out, the neuronal assemblies always turn out to compute something else as well, in addition to the alleged time tracking. For example, in monkey parietal cortex, neuronal spikes are assumed to be related to the judgment of time. At the same time, these neurons also compute velocity and acceleration, measure distance,

55. Pastalkova et al. (2008); Fujisawa et al. (2008); Itskov et al. (2011).

56. Of course, the proper term should be "duration cells" because cell assemblies track duration rather than absolute time.

57. Eichenbaum (2014); MacDonald et al. (2011); Kraus et al. (2015); Tiganj et al. (2017). However, once velocity is monitored at high resolution, the distance and duration of a run are naturally identical (Redish et al., 2000; Geisler et al., 2007; Rangel et al., 2015).

code for spatial attention, plan movements, and prepare for decisions. Thus, these labels largely reflect the perspectives and questions of the experimenter rather than segregated streams of spike information for downstream readers. Indeed, computational models of decision-making use virtually identical logic to the ramping accumulator-threshold models of time-interval mechanisms.[58] Again and again, we find that duration calculations are linked to something else.

In summary, sequential neuronal activity can track the succession of events, which can be placed on a timeline when compared to units of a clock. However, they do not show that neuronal activity actually computes either clock time or duration. It is not yet clear whether neuronal circuits anywhere in the brain are dedicated to computing time as an independent function or, alternatively, whether circuits compute specific functions that evolve sequentially and therefore correlate with the ticks of an external instrument.

Time Warping

Albert Einstein noted that time flies when we in the company of pleasant people, and it slows when we are bored.[59] Experiments support this intuition. Highly motivated states, uncertainty, novel situations, and focused cognitive activity are associated with underestimation of time. Conversely, fearful or aversive situations, fatigue, and sleepiness are associated with overestimation of time. Brain-arousing drugs can accelerate, while sedating drugs and substances decelerate or distort, our subjective sense of time. Drugs affecting dopaminergic signaling prominently modulate timing behavior, potentially through their impact on the basal ganglia, a structure that is critical in time-interval estimation. Dopamine-producing neurons in the substantia nigra and ventral tegmental area become activated around the time when reward is expected and

58. Accumulation of "evidence" and reaching a threshold, determined by comparison to previous knowledge, is the act of decision (Leon and Shadlen, 2003; Janssen and Shadlen, 2005). Accumulation of "something" is calibrated against clock units and can be called time (Machens et al., 2005; Lebedev et al., 2008; Finnerty et al., 2015). An example of such a process is the accumulation of electric charge (evidence) in neurons, brought about by numerous excitatory postsynaptic potentials and culminating with the action potential at a threshold. From an anthropomorphic view, this is a decision made by the neuron. During the delay period of conditioning, increasing population activity can be viewed as a time command (McCormick and Thompson, 1984; Thompson, 2005). Alternatively, one can interpret the same accumulation of spiking activity as preparation for the response.

59. The brain state also affects subjective estimation of distance. When we are engaged in deep thought or a conversation with someone, long walks appear short (Falk and Bindra, 1954). See also Jafarpour and Spiers (2017).

decrease their spiking when the expected reward is omitted or when the expected time of reward delivery varies.[60] Transient selective activation of dopaminergic neurons in mice is sufficient to slow down the judgment of duration, while their inhibition has the opposite effect.[61]

There is a difference between "feeling the moment" and "remembering the moment." Subjective time-bending goes in the opposite direction during feeling versus remembering. When I am giving my 30-minute talk and the moderator politely stops me at 40 minutes, I am in disbelief. Time feels so fast. But when I recall the talk, I wonder why the few things I wanted to convey did not fit into 30 minutes.[62]

Our brain can cheat us not only in judgments of duration but also in judgments of time points. Test this yourself by touching your nose with your toe. They seem to feel the contact at the same time, right? Your eyes saw your nose and toe touching simultaneously, and this event is registered by the visual system. On the other hand, touch information from the toe takes several times longer to reach the brain via the spinal cord than does information from the nose, which is just next door to the brain. Thus, signals from the events do not arrive at the same time in the somatosensory cortex. Yet the brain, the overall observer, learns to compensate for such delay differences and to incorporate them into its computation.[63] Similarly, when we snap our fingers, the somatosensory, visual, and acoustic information blends into a simultaneous event, even though signals of these modalities reach the brain at different times. Experience-driven time-prediction in the brain compensates for such physical difference. However, when the brain is not prepared to compensate, bizarre

60. The discovery of the relationship between dopamine signaling and reward expectancy (Hollermann and Schultz, 1998; Schultz, 2015) gave rise to the burgeoning field of reinforcement learning and machine learning (Sutton and Barto, 1998) and recently linked neuroscience with economics and game theory (Glimcher et al., 2008).

61. Honma et al. (2016). Direct optogenetic manipulations of the dopaminergic neurons not only affect the postulated internal clock but also motor behaviors as well, posing the question of whether the delayed or premature responses reflect fundamental motor effects or internal timekeeper mechanisms (Soares et al., 2016).

62. Judging duration and feeling duration are different things (Wearden, 2015).

63. The physicist would say that this is simply an issue of the reference frame. If two events occur at different locations, they can be judged simultaneous in one reference frame or delayed to each other in another reference frame. An observer standing halfway between two distant flashes in a thunderstorm would see the flashes as simultaneous, while another observer standing closer to one than to the other would see the closest first, followed by the more distant. Note that the latter situation is exactly analogous to the relationship between nose, toe, and brain. However, the brain can learn about such delays through action and edit the delays by computation.

temporal illusions (Chapter 3) can occur. For example, participants in an experiment were instructed to make things appear on a computer screen by pressing a key. Unknown to the participants, the experimenter then introduced various delays between the key presses and the screen events. The brain learns to compensate for such delays, and the cause–effect feeling persists. After this adaptation training, however, the normal relationship between key presses and screen events returned to normal, participants reported that it felt as if the screen events were causing the key presses.[64] Their perception of time had reversed.

Subjective time compression happens with every saccadic eye movement, although we do not notice it. Vision is typically thought of as a space-exploring mechanism, but it also affects timing. When the eyeballs are jumping, the duration of a visual stimulus is underestimated by approximately a factor of two. A 100-ms stimulus around the time of a saccade is judged to be the same duration as a 50-ms stimulus presented to a stationary eye. This time compression is specific to the neuronal mechanisms that support saccadic eye movements because it does not occur during blinking or when auditory clicks, instead of flashes, are used for duration judgment. Remarkably, if research participants are asked to judge the temporal order of events, their performance deteriorates in a small window (–70 to –30 ms) before the saccade, and often their order judgments are reversed so that events that occurred second are perceived to occur first.[65] The saccade-induced time compression is tantamount to a slowdown of a hypothetical neural timing mechanism on which judgments of subjective time depend.

Here is another everyday example of a false reconstruction of the order of events. You are driving on a highway, and a deer crosses the road. You slam on the brakes. This is the sequence of events that you believe happened: you noticed a deer and, as a result, pushed the brakes and turned the car. In reality, your motor reaction time is much faster than the time required for awareness. You slammed on the brakes before consciously noticing the deer. Just as in the toe-nose touch situation and the saccade-induced time reversion, the brain

64. Eagleman et al. (2005); Yarrow et al. (2001). These observations are the modern-day replication of the findings by William Grey Walter in patients with implanted electrodes in the motor cortex, which he presented to the Ostler Society, Oxford University (1963; see Dennett and Kinsbourne, 1992). In Grey Walter's experiment, the patients advanced slides of a slide projector at will by pressing the controller button of the projector. However, the controller in fact was not connected to the projector. Instead, the CNV signal from the motor cortex of the subject was the trigger. Since the CNV builds up tens to hundred milliseconds before action, the carousal often started to move before the actual button press. The subjects were startled and felt that an agent was acting on their behalf when they were about to push the button.

65. Morrone et al. (2005). These movement-induced, time-warping illusions are reminiscent of the "time traveler" problem of relativity.

retrospectively constructs temporal order and causality. This happens because brain-constructed (subjective) time is relativistic; it can flow forward or backward. Copies of the automatic, short-latency actions from the brainstem are forwarded to cortical areas, where such information is interpreted. This neuronal interpretation gives rise to the internally generated prediction that "I am the agent of this action." In turn, watching yourself produces an illusion of temporal order and cause-and-effect relationship.[66]

To summarize, time-estimation experiments show that subjective time can be warped by many manipulations. Einstein was right: duration is relative. In addition, these observations demonstrate that the brain does not faithfully emulate the passage of physical or clock time. Time mechanisms in the brain run at multiple scales and manifest relatively independently in different structures in the service of their own computations. Most importantly, no experiment shows convincingly that there are dedicated mechanisms and neurons whose sole function is time generation or counting the number of oscillation cycles.[67] While all neuronal computation evolves on a time line, subjective time can move either forward or backward.

SPACE AND TIME IN THE BRAIN

Space and time have deep conceptual similarities and they share relational structure. Spatial schemas can be used equally well for temporal schemas to organize events in sequential order. Distance and duration appear to be represented in the brain by the same structures and likely by the same neurons

66. This may explain why we experience the urge to move a finger, for example, several hundred milliseconds after the onset of a readiness potential (CNV) above the supplementary motor area, a brain region believed to be critical in voluntary action. Neuronal "interpretation" needs approximately 500 ms to become available for awareness (Libet et al., 1979). An interesting question is whether we notice the feeling of the urge to move if we are not actively attending to the action. At this moment, you do not feel that you have a shirt on. But after reading this sentence, you might. The internal interpretation can be viewed as "attention." In turn, the act of attention and the underlying neuronal mechanisms can be regarded as the source of conscious experience (Graziano, 2013).

67. Neurons in the hypothalamic suprachiasmatic nucleus of the mammalian brain are portrayed as dedicated timekeepers that oscillate with an internal period of approximately 24.2 hours and maintain a stable phase relationship with solar time. Such a circadian clock is present in virtually all terrestrial and most aquatic animals and enables them to coordinate their behavior with the day–night cycle. Such recurrences and environmental regularities allow organisms to reliably predict the consequences of their actions, whether we call them timing behaviors or not. By this logic, the rotation of the earth is also a timekeeper because it changes cyclically.

and mechanisms. The preceding physiological data show that duration computation relates to distance computation. If these computations in the brain are related, as they are in physics, they should co-vary.

Imagine that our friends come over to watch a movie. After a while, we stop the movie and ask everyone to mark on a rod how much time has passed. Will the markings be similar on an identical length rod? What if everyone gets a rod of different length? My guess is that everyone's marking will reflect approximately the same ratio, independent of the rod length. Furthermore, nobody will argue that, "hey, we are measuring distance not time."[68] Analogous to egocentric and allocentric view of spatial relationships (Chapters 5 and 7), we also conceptualize time by such dual measures: we move through time or it passes us. In the egocentric view, the observer is moving on a timeline ("We are approaching the deadline of grant submission"). The future is in front of us. In contrast, in the allocentric metaphor, the observer is stationary and time is a river which carries things relative to the observer ("The deadline of grant submission is approaching"). When two objects are moving, their temporal relationship is based on their direction of motion (*A* is ahead of *B* or vice versa). Because of the duality of the event sequencing nature of conceptual time, we often get confused: "Should I turn my watch forward or back six hours when flying from New York to Europe?"[69]

Feeling duration depends not only on mood but on many other factors. When two objects are presented separately but for the same length of time, the duration of the larger object feels longer. Such subjective time expansion also occurs when a low-probability "oddball" event is embedded in a familiar sequence of stimuli. The novel stimulus feels longer than the familiar stimuli. Similarly, the first stimulus in a repeated train is judged to have longer duration than successive stimuli.[70] Subjective time moves at a different pace than clock time.

An elegant demonstration of the space-time similarity of subjective experience comes from research in architecture. In that experiment, students of architecture were asked to imagine that they were a Lilliputian figure in scale models

68. This experiment shows that our guess is a comparison against something that we have previously calibrated (learned). To guess the elapsed time, we had to have experience with many movies to learn that they typically last 100 minutes. The duration guess would be identical if we asked our friends to mark the pitch of sound along a sweep from, say, 1 kHz to 10 kHz. In each case, we compare the ratios of different modalities. Ratio estimation is easy for the brain, and it is independent of the measuring units. Such relations are universal (see Chapter 12).

69. Lakoff and Johnson (1980) point out that we often use metaphors to talk about abstract domains such as space and time and that we move from concrete domains to the more abstract, similar to the episodic-semantic memory transformation. See also Boroditsky (2000).

70. Tse et al. (2004).

that were 1/6, 1/12, or 1/24 of their full-size lounge on campus and to engage in the activities that they usually do there. The miniature lounges were furnished with chipboard furniture and proportionally sized figurines. The key parameter of the experiment was subjective time: the participants were asked to inform the investigator when they felt that 30 minutes had passed. The participants found that subjective time accelerated. Even more surprisingly, the magnitude of the acceleration, measured as the ratio of experiential duration to clock time, was quantitatively predicted from the proportions of the models relative to the full-sized environment. For example, if a participant experienced 30 minutes in a 1/6 scale environment in 10 minutes of clock time, the subjective 30 minutes in a 1/12 model occurred in just 5 minutes. This experiment demonstrates that temporal and spatial scales are related in subjective experience and that spatial scale may be a principal mediator of experiential time. Such space-time relativity implies that the experience of space and time refers to the same thing.[71] Experiencing time is experiencing space.

Space and Time Warping by Action

A number of neuronal mechanisms compensate for the disturbing sense of motion during saccadic eye movements. Under rare conditions, these useful mechanisms produce illusions. Saccades not only affect temporal judgments but also distort space. Objects flashed briefly just before and after the beginning of the saccade appear to be in false positions. They seem to move in the direction of the saccade target, and they are squashed parallel to the path of the saccade. As a result, geometric relationships between objects get distorted temporarily. Such warping of space and time is not a direct consequence of the fast-moving eyes because the distortion starts tens of milliseconds before and lasts until after the saccade. Instead, it reflects the corollary discharge mechanism (Chapter 3) that keeps track of motor commands.

Neuronal recordings also support the equivalence of perceived distance and duration. Neurons in the lateral intraparietal cortical area of the monkey appear to encode temporal duration as well as spatial distance. Approximately

71. DeLong (1981) performed these experiments in the School of Architecture, University of Tennessee. Spatial scale was based on the linear dimension, not volume. This suggests that, for humans, two-dimensional scaling is related to duration. The large number of subjects allowed proper quantification of the findings and formulation of the relationship as $E = x$, where E is the experiential duration and x is the reciprocal of the scale of the environment being observed. See also Bonasia et al. (2016).

one-third of the neurons in this brain region change their coding predictively during saccadic eye movements, and their activity is believed to be responsible for the perceived compression of space and time.[72]

In a locomotion experiment, normal participants were asked to reproduce a previously walked straight path while blindfolded. Although the velocity was reproduced accurately, the walking duration was affected to the same degree as the distance, suggesting that brain computation of space and time are closely related. In a more sophisticated version of this experiment, each participant was passively transported forward in a robot from 2 to 10 meters while blindfolded and head-restrained. After the robot came to a complete stop, the participant drove the robot with the joystick in the same direction in an attempt to reproduce the distance previously imposed. Not only the sample passive distances but also their velocity profiles were varied in a random sequence of triangular, square, and trapezoidal patterns, yet the participants reproduced them reliably. The results demonstrate that humans can reproduce a simple trajectory using only vestibular and possibly somatosensory cues and can store dynamic properties of whole-body passive linear motion without seeing the world.[73]

Examining the distance–duration issue makes us wonder about the role of velocity on neuronal firing. When a rat runs through the place field of a place cell at different velocities in the maze, the number of spikes emitted in different segments of the field remain largely the same (Figure 7.5). Moreover, the oscillation frequency of individual place cell spikes varies as a function of the rat's running velocity, and, for this reason, they are called *velocity-controlled oscillators*.[74] From the elapsed duration measured by the time-tracking ability of assembly sequences within the place field and instantaneous velocity calculated

72. Not only space and time but also numbers are distorted by saccades (Burr and Morrone, 2010), leading to the suggestion that parietal neurons code for magnitude rather than for time or space (Walsh, 2003). A related view is that several types of cognitive tasks, such as mental arithmetic, spatial and nonspatial working memory, attention control, conceptual reasoning, and timekeeping recruit similar structures in imaging experiments; therefore, the underlying computation may be similar, perhaps described by the term "cognitive effort" (Radua et al., 2014). Consult also Basso et al. (1996); Glasauer et al. (2007); Cohen Kardosh et al. (2012).

73. Berthoz et al. (1995); Perbal et al. (2003); Elsinger et al. (2003). Parkinsonian patients strongly underestimate intervals above 10 s on time-production tasks compared to controls, and the time underestimation correlates with the level of striatal dopamine transporter located on presynaptic nerve endings (Honma et al., 2016). Patients with parietal lesions not only suffer from spatial neglect but also grossly under- or overestimate multisecond intervals.

74. This term is a deliberate reference to the voltage-controlled oscillators in electronics; in these devices, the output frequency is relatively linearly controlled by a voltage input (Geisler et al., 2007; Jeewajee et al., 2008). We used the term "speed" rather than "velocity" because identical observations were made during wheel running in a memory task without translocation.

from the oscillation frequency of the place cells, the travel distance and instantaneous position can be continuously derived. How the brain performs such calculations remains to be learned. But we have some clues.

As in physics, displacement and duration are two sides of the same coin in the brain and are linked by velocity. This relationship explains why most hippocampal neurons report distance and duration equally well. If the animal runs a very long distance or duration, every hippocampal neuron will become a place (distance) cell and will also report duration via its position in the neuronal trajectory sequence.[75] If the same neurons function as both place cells and time cells, then downstream reader mechanisms have no way to tell whether the spikes represent distance or duration. Short of a demonstration of separate downstream reader mechanisms of distance and duration, we can conclude that hippocampal cells are neither place nor time cells.[76]

DISTANCE–DURATION UNITY AND SUCCESSION OF EVENTS

The dashboard of my Lotus Elise has multiple displays, including an odometer, a speedometer, and a clock. The odometer keeps track of the distance the car travels. This measure is essentially the number of strokes of the piston in the engine's cylinder.[77] My lab is approximately 1.5 million strokes from home. Because this number is not a very informative measure beyond the Lotus's world, one can relate the number of strokes to the circumference of the tires. In turn, the circumference can be related to some agreed unit and expressed as inches, feet, yards, meters, miles, light years, or whatever units one chooses and displayed on the odometer. Hence, we derived distance from the number

75. The small fraction of neurons that appear to be differentially sensitive to either duration or distance may reflect a wide distribution of velocity-dependence within the neuronal population or a potential error in measuring the instantaneous speed (velocity) of the animal (Kraus et al., 2015).

76. During treadmill running in a spatial memory task, sequential firing of entorhinal cortical neurons can also keep track of elapsed time and distance. Grid cells are more sharply tuned to duration and distance than are non-grid cells. Grid cells typically exhibit multiple firing fields during treadmill running, similar to their periodic firing fields observed in open fields, suggesting a common, rather than separate, mechanism for processing distance and duration (Kraus et al., 2015).

77. This is a super-simplified description of the engine's working, of course. A four-stroke cycle engine completes five strokes in one operating cycle, including intake, compression, ignition, power, and exhaust strokes. The piston movements in the 4, 6, or 8 cylinders need to be coordinated.

of strokes. The speedometer uses a small magnet attached to the car's rotating drive shaft and a coil. Each time the magnet passes the coil (the sensor), it generates a bit of electric current and the magnitude of the electric current is converted to units on the speedometer's display. Older mechanical versions use the same conversion principle. Ultimately, the displayed instantaneous reading on the speedometer is related to the rate of the movement of the piston in the engine's cylinder. The faster the piston moves up and down, the faster the car runs. If we divide the 1.5 million strokes by the average readings of the speedometer, we can tell how long it takes for me to get to work. Relating this ratio to some agreed units, such as ticks of a clock, the amount of sand flowing through a sand clock, heat beats or other relational units, we can call it duration or time. The relationship between the piston movement-generated "time" and time on the dashboard clock may vary depending on the "subjective mood" of the car. Using a higher octane gas makes my car happier (i.e., it runs faster) as is also the case when I press the accelerator harder and supply more gas.

The important wisdom of my car metaphor is that the very same mechanical movement (the piston's movement and its rate of movement) can be interpreted as speed, distance or duration, as long as we relate the mechanical movement to units of human-invented measuring instruments. But the car itself does not make either distance or duration, even if the piston and the other moving parts of the engine have the same mechanism as many mechanical clocks.

Neuroscientists search for the dashboard of the brain with a speedometer, odometer, and a clock. But let's face it: despite their seductive allure, we have no direct experimental evidence for neuronal mechanisms dedicated to selectively and specifically compute either space or time, although we have hard evidence for the existence of a speedometer (the vestibular system, muscle reafferentation, optic flow, etc). There is no such a thing as "perception of time" or "perception of space"—they are simply titles in James's list (Chapter 1).[78] Perhaps they are only grand abstractions of the human mind and we actually do not need them to describe brain operations. We need the space and time metaphors of classical physics so that that our mind can travel freely and confidently within them.

Imagine a collection of adjacent cups into which arbitrary items or events can be placed. A walk from event to event then can be described as a journey through space. If the duration of progression from one cup to the next is

78. This turns out to be a 2,000-year-old idea: "time by itself does not exist. . . . It must not be claimed that anyone can sense time apart from the movement of things or their restful immobility" (Lucretius, 2008, Book 1). We assume that time exists, but we never observe it directly. Instead, we deduce it from counting events.

considered, it is a trip through time.[79] If the rates of the transitions from cup to cup are known (velocity), the journey in time is identical to the journey in space because the distances between cups can be deduced. Conversely, if the distances between the cups are combined with velocity information, the journey through space is equivalent to the trip duration. Alternatively, we can just call the journey a trajectory or a sequence with transitions. This simplistic description has the advantage of creating a one-to-one relationship between sequences in the outside world and neuronal trajectories in the brain without the need of referring to units of human-made tools. As discussed in Chapters 7 and 8, brain dynamics generate extraordinarily large numbers of such trajectories, which can be split or concatenated in a number of ways, thus creating versatile opportunities to match learned experience with trajectories. These flexible mechanisms ensure that higher order connections can also be established without any need to refer to time. Time can always be explained in nontemporal physical terms. The measurement of time by precise instruments simply provides a practical imaginary reference so that a succession of events in two or more systems can be compared.[80]

Of course, the reader might say, "OK, change is just another name for time." However, there is a fundamental difference. Change always refers to something ("change of something"). In contrast, time is defined as independent from everything else. Together with space, it is this alleged independence from everything else that gives them such powerful conceptual organizational powers.

Motion, Time, and Space

Can we define progression of events in the brain from first principles? Navigation in real or mental space corresponds to the succession of events. In the most general sense, it is a motion. As defined in physics, motion is relative (between at least two bodies) and always characterized by velocity.

79. Theorists may call it a *saddle point attractor-repellor state vector*. Change can be continuous or discrete. A drive on an ideal flat highway is continuous. Most country roads go through hills and valleys, relatively discretizing the drive. In conceptual space, such hills and valleys can be called "attractors" and "repellors," which can change dynamically and determine the rate of change of transitions.

80. Perhaps Karl Lashley (1951) should be credited with this insight. He thought about these issues before dynamical theory gained prominence: "memory appears almost invariably as a temporal sequence, either as a succession of words or of acts. . . . Spatial and temporal order thus appear to be almost completely interchangeable in cerebral action. The translation from the spatial distribution of memory traces to temporal sequence seems to be a fundamental aspect of the problem of serial order."

Although we have no direct sensors for space or time,[81] we do have vestibular sensors to detect changes in velocity (i.e., acceleration) and the head direction system can calculate vectorial displacement. Motion per se can be directly experienced from optic or tactile flow and corollary discharge without the need to sense space or time.[82].

Hippocampus as a Sequence Generator

Navigation in real or mental space is, by its nature, a succession of events. Perhaps that is all the hippocampal system does: encode content-free or content-limited ordinal structure, without the details of particular events. Indeed, a parsimonious interpretation of all the experiments discussed in this chapter is that both the parietal cortex and the hippocampal-entorhinal system are general-purpose sequence generators that continuously tile the gap between events to be linked.[83]

Let me support this heretical statement with an experiment that illustrates the contrast between sequences and apparent space or time representation. A group of patients with damage to the hippocampus and some normal individuals received a guided tour of the campus of the University of California (Figure 10.2). The tour included various events, preplanned by the experimenters, such as discarding a cup, locking a bike, buying a banana, finding a coin, drinking from a water fountain, and so forth. At the end of the trip, the participants returned to the laboratory, where they completed three tests. The first one asked the participants to describe everything they remembered about the walk in six minutes. In the second test, the experimenter provided a prompt for each of the events ("What happened to the bicycle?") and asked the participants to provide

81. The circadian clock of the suprachiasmatic nucleus is perhaps closest to a man-made clock in the sense that it oscillates "around the clock." However, oscillations and time are orthogonal dimensions (phase vs. duration). Events in an oscillator always return to the same zero phase, quite different from the infinite time arrow idea. Phase can be reset, advanced, or delayed. Time just moves forward. The circadian clocks in the brain, autonomous nervous system, liver, and gut runs spontaneously and largely independently of each other, but they can be reset and synchronized by light, hunger, and other variables, which have sensors to mediate such inputs.

82. After ambulatory calibration of the distance metric, egocentric distance may be extracted from the spiking of binocular disparity-selective neurons in the primary visual and the posterior parietal cortex (Pouget and Sejnowski, 1994). Optic flow is critical for grid cells and spatial metrics (Chen et al., 2016).

83. Of course, many other brain structures can generate sequences internally. However, the hippocampus, as a giant cortical module, is a special case since its sequences can affect neuronal content in the large cortical mantle.

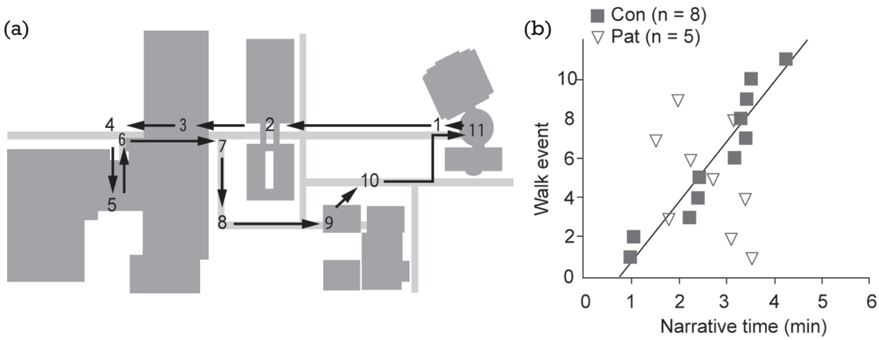

Figure 10.2. The core deficit in patients with hippocampal lesion is their inability to narrate events in the order they occurred. A: Map of eleven events that occurred during a guided campus walk. Sidewalks are light gray. Buildings are dark gray. Arrows indicate the path taken during the walk. B: Events from the walk, described during 6-minute narratives. The control group (*grey squares; Con*) tended to describe the eleven events in the order that they occurred. The order in which the patients (*open triangles; Pat*) described events was unrelated to the order in which the events occurred. Reproduced from Dede et al. (2016).

all the details they could recall in one minute. In the third test, forced-choice questions were asked about the events ("Did we find a quarter or a dime?"). The normal participants also took the same tests again one month later.

As expected, the hippocampus-damaged patients were worse than the normal control participants on most tests. The patients (tested immediately after the tour) performed about as well as the control participants tested one month later on measures like the number of events remembered and duration judgments (except for one participant, who had the greatest brain damage, possibly involving the hippocampus, parahippocampal areas, and frontal cortex). Despite their memory impairment, hippocampal patients recalled many spatial and perceptual details, such as that the bicycle "had a light in the front." The events that were most memorable for the control participants were also most memorable for the patients. Yet there was a striking difference between the two groups. The order in which patients recalled the events was unrelated to the order in which they occurred. In contrast, control participants effectively recalled the order of the events even a month later.[84]

Let me cite another real-life example. A London taxi driver who had sustained bilateral hippocampal damage retained considerable knowledge about city's landmarks and their approximate locations, but he could not compile those

84. Dede et al. (2016). For functional magnetic resonance imaging support of sequence coding, see DuBrow and Davachi (2013, 2016).

locations into a novel sequential trajectory. Rats with hippocampal damage showed similarly impaired memory for the order of odor stimuli, in contrast to their preserved capacity to recognize odors that had recently occurred.[85] These experiments support the idea that the hippocampus is a neuronal sequence generator that can create a nominal order of sequential events and reproduce it upon recall.

Whether in reference to space or time, sequences in the hippocampal system may simply point to the items (*what*) stored in the neocortex in the same order as they were experienced during learning. This division of labor is analogous to the role of a librarian (hippocampus, pointing to the items) in a library (neocortex, where semantic knowledge is stored).[86] Thus, the hippocampal system may be responsible for constructing sequences of information chunks, while chunk content is encoded in and retrieved from the neocortex.

The concept of the hippocampus as a sequence generator has the advantages of separately encoding items and events in hypothetical space and time, as discussed at the beginning of this chapter. Instead of storing a huge repertoire of unique states for every sequence encountered, the brain can store sequences (in the hippocampus) separately from the content (neocortex). Instead of storing every possible neuronal sentence, the cortex can separately store all the words, while the hippocampus concatenates their sequential order. This organized access (embedded in neuronal trajectories) to neocortical representations (*what*) then becomes episodic information.[87] Episodic memory is then an ordered sequence of "*whats*."

85. Fortin et al. (2002); Maguire et al. (2006).

86. Teyler and DiScenna (1986) used the indexing metaphor for hippocampal computation. The index or table of contents (hippocampus) represents a concise summary of a book. However, instead of a single index that "points" to cortical modules, the "unit" is a sequence of indices (a "multiplexed pointer") such that the hippocampal system is responsible for concatenating neocortical information chunks into sequences for both encoding and retrieval, typically in theta oscillation cycles (Buzsáki and Tingley, 2018).

87. Karl Friston and I gave talks back to back at a meeting in the gorgeous building of the Hungarian Academy of Sciences in January 2016. To the audience, it felt as if we lived in two different worlds (he talked about the Bayesian brain, while I talked about internally generated sequences and log distributions; see Chapter 12). To us, it seemed that our thoughts resonated, and we wanted to discuss the potential links over lunch. After several hours of friendly debate, we left the table with a basically finished paper in our heads about the advantages of sequential order generation. Karl did the heavy lifting of writing most of the manuscript (Friston and Buzsáki, 2016). This lunch discussion was also an opportunity for me to observe how the brain of a genius works.

Memory for the Future

Neuronal trajectories and their ordering-pointing roles can equally well represent the past, present, or future. Predicting the future is possible only after we have experienced the consequences of our actions and stored the outcomes of both successful and failed attempts. We might expect, therefore, that the neuronal mechanisms of postdiction and prediction are not that different. Indeed, imaging experiments in humans show that many structures that were traditionally viewed as part of memory systems are inseparable parts of planning, imagining, and action systems.[88] In laboratory animals, what appears to be a memory of the past in the sequential activity of cell assembly sequences can also reflect planned future action. Perhaps representation via ordered neuronal sequences explains why the brain structures and mechanisms associated with memory and planning show such a strong overlap. The distinction between postdiction and prediction may be analogous to the illusory separation between the past, present, and future in physics.

With or Without Space and Time?

Do not get me wrong. I am not suggesting that neuroscience researchers dispense with the concepts of space and time. The formulation of spacetime by physics did not discredit clocks. Quite to the contrary, physics and all other sciences rely increasingly more on the precision of measuring instruments to calibrate experimental observations. This is not expected to change in neuroscience either. Indeed, in the remainder of this book, I continue to use these terms in the usual manner, with the understanding that they make sense only

88. Planning could be referred to as "constructive" memory (Schacter and Addis, 2007) or "memory of the future" (Ingvar, 1985). See also Lundh, 1983; Schacter et al. (2007); Hassabis and Maguire (2007); Buckner and Carroll (2007); Suddendorf and Corballis (2007); Pastalkova et al. (2008); Fujisawa et al. (2008); Lisman and Redish (2009); Buckner (2010); Gershman (2017). These new findings are beginning to bring down the walls that have long stood between concocted words and brain mechanisms. Planning and recalling feel different: one relates to the subjective future, the other to the subjective past. Yet any form of planning requires access to preexisting knowledge (Luria, 1966). Also, a planned action cannot be carried out unless the "plan" is placed into memory until its actions are completed. Bilateral hippocampal damage is associated not only in recalling the past but also with an impairment in thinking about one's personal future (Atance and O'Neill, 2001) and in imagining new fictitious experiences (Hassabis et al., 2007).

in comparison with units of instruments. This should not be taken as an inconsistency but as a way not confusing the average reader. Time and space remain useful concepts for organizing the thought.

If no radical change in practice is recommended, why spend so many pages scrutinizing space and time representations in the brain ? The answer is that the observation that the evolution of neuronal activity is reliably correlated with the temporal succession of events ("temporal sequence of representations") does not mean that neuronal activity computes time ("representation of temporal sequences"). Neither instruments nor brains create space or time.[89] Yet they remain useful concepts because without them complex thinking would be nearly impossible. Yet, it is useful to remember that it is one thing to compare the evolution of brain activity against clocks but quite another to design a research program to search for neuronal "representations" of space, time, or other mental constructs in the brain.

If "my ideas" appear too radical to the reader, here is a reminder that I am not the first weird person to think this way. The Greek philosopher, Philo of Alexandria wrote these lines two millennia ago: "Time is nothing but the sequence of days and nights, and these things are necessarily connected with the motion of the Sun above and below the Earth. But the Sun is part of the heavens, so that Time must be recognized as something posterior to this World. So it would be correct to say not that the World has created Time, but that time owed its existence to the World. For it is the motion of the heavens that determines the nature of Time."

The inside-out approach to space and time may be generalized to other terms listed in William James's list, nearly all of which may exist only in our mind. Our challenge as neuroscientists is how to perform experiments and think about brain mechanisms without resorting to such preconceived ideas.

I left out one thing from the discussion about the relationship between sequences and duration: the rate of succession. Only when sequential order is combined with the rate of change can it replace time. The pace of transition in the brain is supported by ubiquitous brain mechanisms known collectively as "gain control," which I address in the next chapter.

89. Bergson (1922/1999); Dennett and Kinsbourne (1992); Ward (2002); Friston and Buzsáki (2016); Shilnikov and Maurer (2016). Ever since Galileo, time has been critical in many equations of classic physics (and in neuroscience; e.g., dv/dt). Yet time is conspicuously missing from equations of contemporary theories. These theories are all about change and relations. Perhaps when a mathematical brain theory eventually emerges, its equations will not use *t*.

SUMMARY

In this chapter, I used space and time to illustrate that even these seemingly non-negotiable terms can be viewed differently from a brain-centered perspective. In both Newtonian physics and Kant's philosophy, space and time are independent from each other and from everything else. Everything that exists occurs in space, assumed to be a huge container, and on a time line, assumed to be an arrow. Newton could imagine an empty world, but not a world without space and time. Yet space is not a huge theater into which we can place things, except in our minds. We can infer space and time only from things that move.

The science of space and time began with the invention of measuring instruments, which converted these dimensionless concepts into distance and duration with precise units. This process created a special problem for neuroscience. If space and time correspond to their measured variants, we may wonder what space and time mean without such instruments, including for non-human animals that cannot read those instruments.

Contemporary neuroscience still lives within the framework of the classical physics view. Our episodic memories are defined as "what happened to me, where, and when." This definition requires the identification of mechanisms in the brain that store the "what" in independent coordinates of the "where" and "when." This is a typical outside-in approach: assume the concepts and search for their homes in the brain. Accordingly, neuroscience researchers have diligently looked for brain mechanisms that define space and time. Two major hubs have been identified for space: the parietal cortex for egocentric representation and the hippocampal system for allocentric representation. Research on timekeeping has been largely independent from research on space. Initially, it was debated whether the brain has a central clock, similar to that found in computers, or if each structure generates timing for its own needs. The cerebellum and basal ganglia have been favored as central clocks. Recent work emphasizes the parietal cortex and hippocampus-entorhinal system as substrates for time perception and generation. These claims have generated an intense discussion about whether the sole function of some neurons is to deal with space or time or whether they always compute something else, too. Similarly, the main source of controversy about "place cells" versus "time cells" is rooted in the internal contradictions of classical physics. The animal's velocity converts between distance and duration representations, making place cells and time cells equivalent.

The independence of space and time has been debated by linguists and physicists alike. Classical physics revealed that distance and duration are linked through speed: knowledge of any two variables can identify the third, so one is always redundant. Contemporary physics, especially general relativity, produced our current picture of spacetime, the equivalence of space and time, which is different from either space or time. This view teaches us that the world does not *contain* things; it *is* things.

Can we do neuroscience without resorting to these "obsolete" concepts? Alternatively, can we support the philosophical stance that the spacetime of physics and "lived" space and time are qualitatively different? The main problem is that the brain does not have sensors for either space or time—but it has sensors for head direction and velocity. Neither brains nor clocks make time. Similarly, neither brains nor rulers make distance. I suggest that it is safe to continue to do neuroscience as we have done it over the past century, using measuring instruments and relating observations to their units. However, it is not safe to declare that some brain region or mechanism represents space or time or calculates distance or duration.

Everything that we attribute to time in the brain can be accomplished by sequential cell assemblies or neuronal trajectories. Hippocampus-damaged amnesic patients have much less of a problem with estimating and recalling distances and durations than with remembering the sequential order of events.